



## **Wheel–Rail Interaction Challenges and Grinding-Based Rehabilitation on Africa’s Cape Gauge Heavy Haul Network: The NACALA Experience**

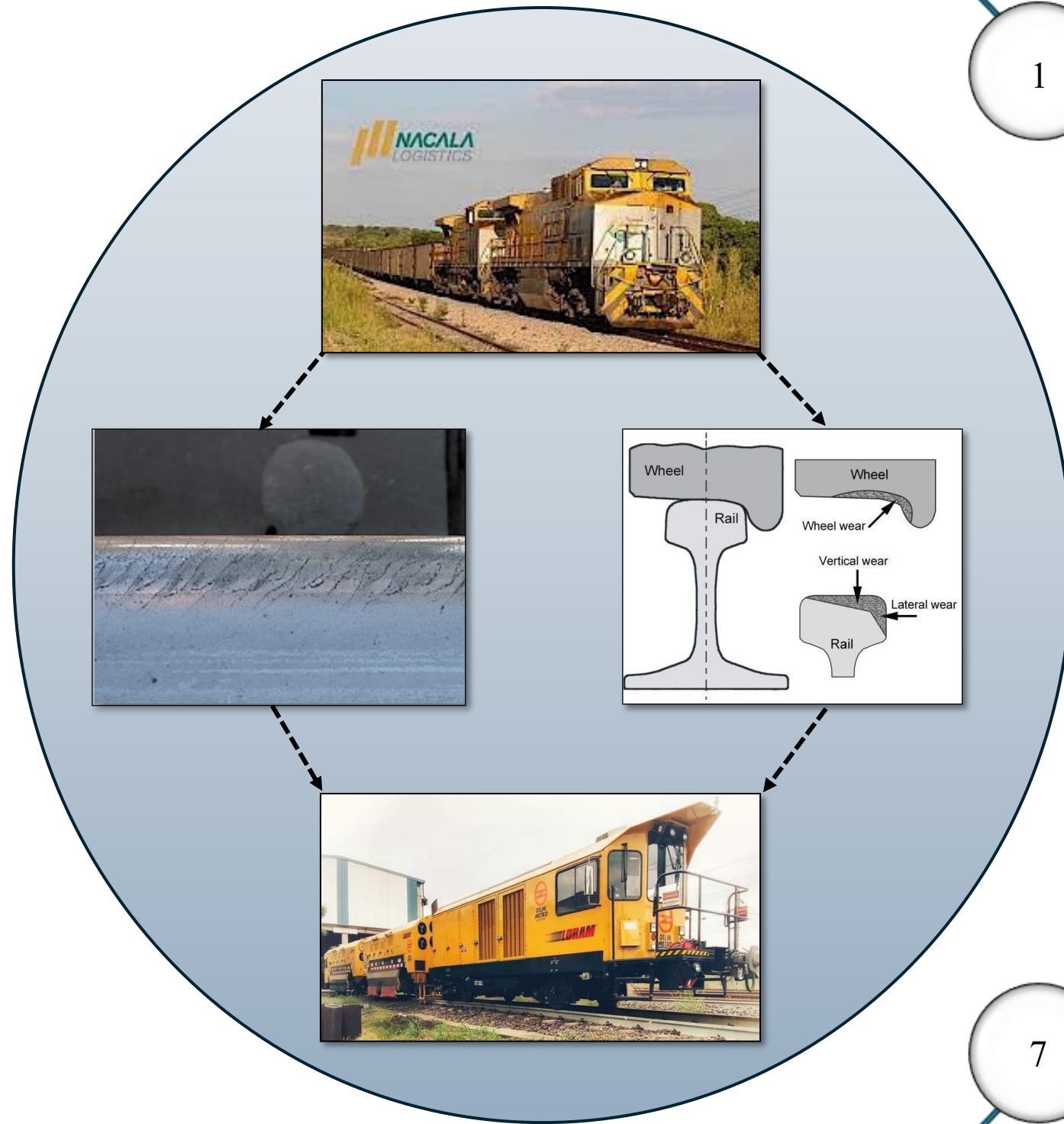
**Mr. B. P. Awasthi, Head & COO, Nacala Logistics, Mozambique**  
**Co-Speaker: Mr. Yash Sehgal, Director, Vandhana International**

DALLAS, TX





# Contents



1

Introduction

2

This Study

3

Methodology

4

Condition of Rail

5

Rail Rehabilitation

6

Case Studies

7

Inferences



# Introduction



DALLAS, TX



# Introduction



DALLAS, TX





# Introduction



|                  |                          |                     |                           |
|------------------|--------------------------|---------------------|---------------------------|
| 18M Tons of Coal | 4M Tons of General Cargo | 1,03,000 Passengers | 30 GMT                    |
| Annually         | Annually                 | Annually            | Expected 40 GMT from 2028 |
| 1067 mm          | AREMA 115 RE             | 22 Ton              | 1650 KM                   |
| Gauge            | Rail                     | Axle Load           | Length                    |



# 115 RE Rail

[illegible]





## Problem Statement

*“With deteriorated rail conditions across key corridors, Nacala Logistics needed to identify the most effective and cost-efficient rail reprofiling strategy to restore wheel-rail performance, control further deterioration and extend rail asset life.”*

*Accordingly, four maintenance pathways were evaluated:*

- *Preventive Rail Grinding*
- *Corrective Rail Grinding*
- *Preventive Gradual Rail Grinding*
- *Rail Milling*





# This Study

## 1. Evolution of rail deterioration under constrained maintenance practices

Rail profiles

Surface condition

DPT testing

Wear on rails

Severity of RCF

## 2. Rail re-profiling based rehabilitation framework that can stabilize wheel-rail dynamics

Design of Templates

Initial Corrective Strategy

RCF Control Strategy

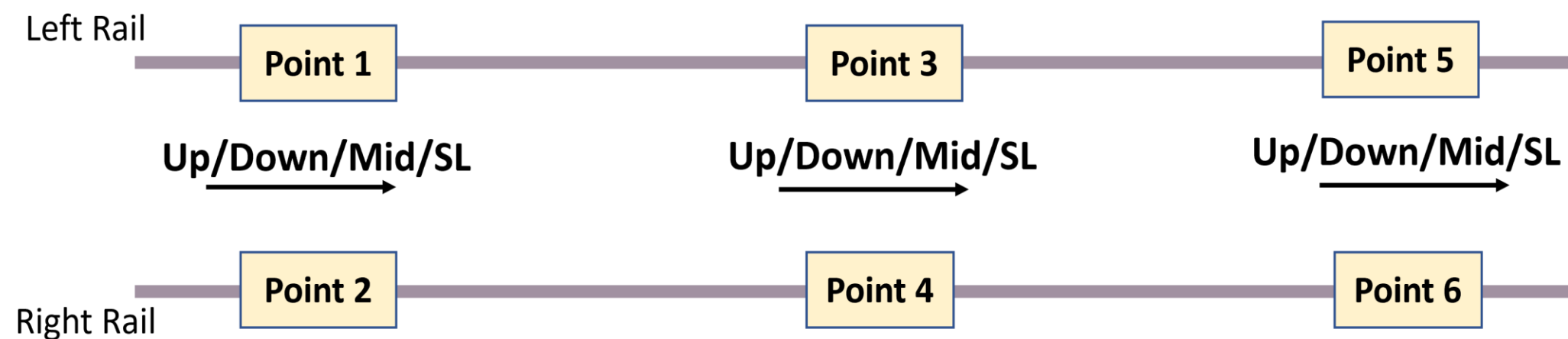
Case Studies

Selected Reprofilng Approach





| Data              | Source                                                                            |
|-------------------|-----------------------------------------------------------------------------------|
| GMT Data          | NLC Operations Team                                                               |
| Rail Profile Data | MiniProf (MP – 262) used for data collection for high accuracy and repeatability. |
| DPT               | Spot-check DPT Kits                                                               |



MP – 262



DPT Kit



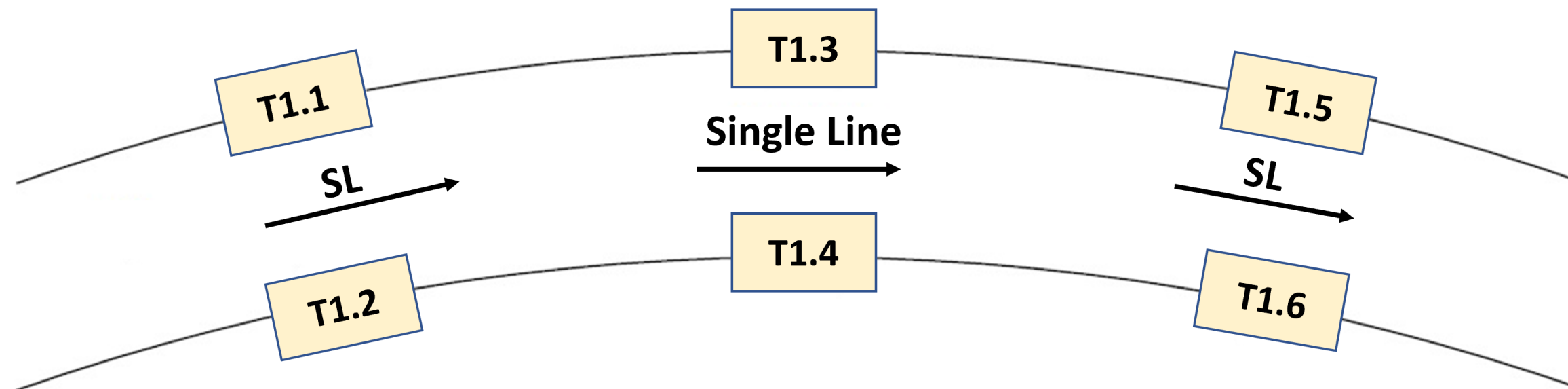
## Test Site: - T01



Sharp Curve - 2.92°

Location: - MONAPO

Date: - 07/11/2024



| Sl. No. | Point Location Details             | Chainage | Sl. No. | Other Key Aspects             | Remarks    |
|---------|------------------------------------|----------|---------|-------------------------------|------------|
| 1.      | T1.1_SECTION 7E_HIGH_SL_07-11-2024 | 65.63    | 1.      | Rail replacement date         | 2015       |
| 2.      | T1.2_SECTION 7E_LOW_SL_07-11-2024  | 65.63    | 2.      | Traffic load/year             | 18 GMT     |
| 3.      | T1.3_SECTION 7E_HIGH_SL_07-11-2024 | 65.78    | 3.      | Traffic load passed till date | 118.37 GMT |
| 4.      | T1.4_SECTION 7E_LOW_SL_07-11-2024  | 65.78    | 4.      | Superelevation                | 20 mm      |
| 5.      | T1.5_SECTION 7E_HIGH_SL_07-11-2024 | 65.93    | 5.      | Rail size                     | 56.74 Kg   |
| 6.      | T1.6_SECTION 7E_LOW_SL_07-11-2024  | 65.93    | 6.      | Length of section             | 547 m      |

THIS IS PROPERTY OF VANDHANA INTERNATIONAL PVT. LTD. UNAUTHORIZED USE COPYING DISCLOSURE, DISSEMINATION OR DISTRIBUTION IS STRICTLY PROHIBITED.  
Copyright© 2021-Vandhana International Pvt. Ltd. All Rights Reserved





# Methodology

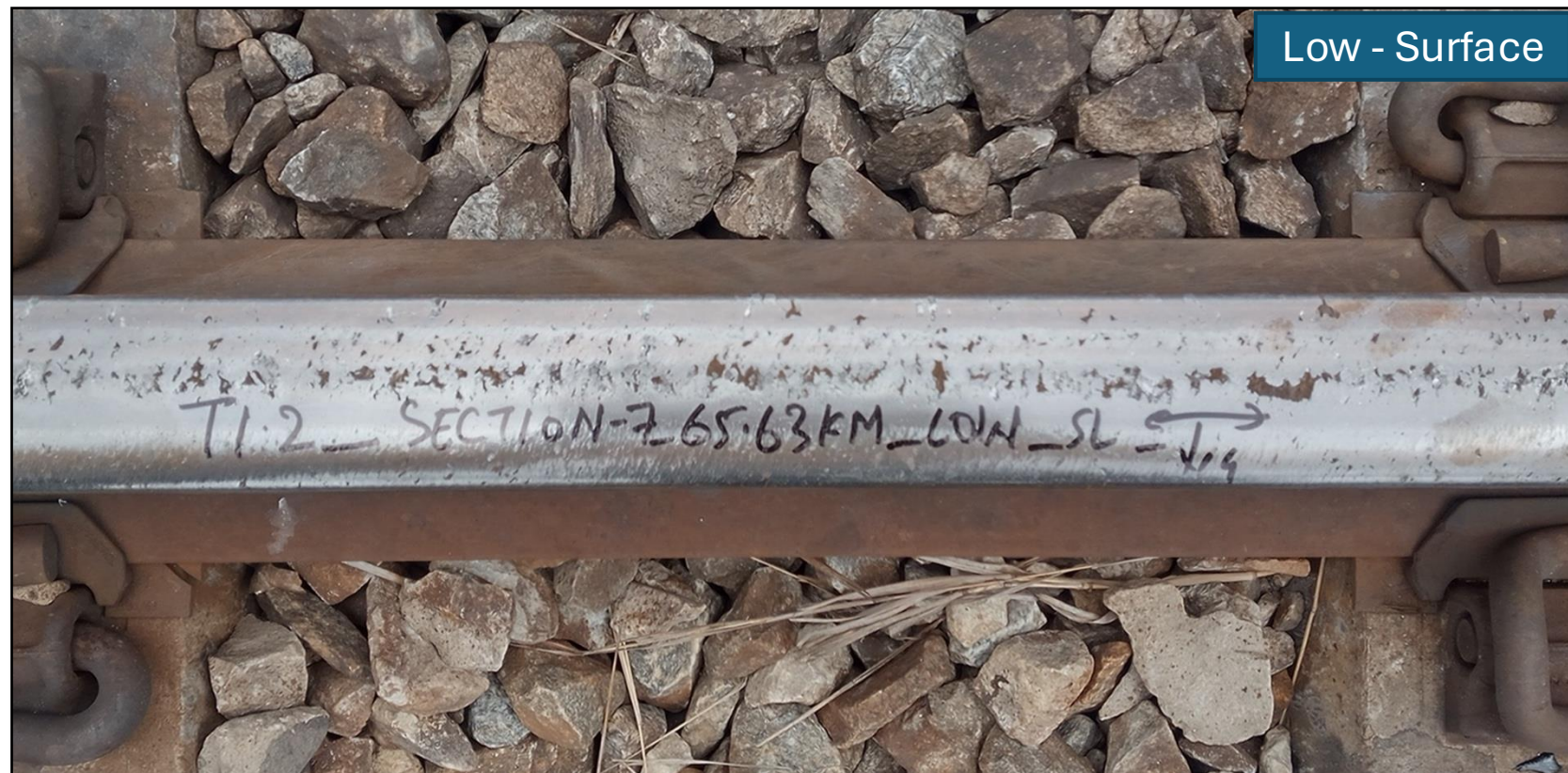


| Test Point No | Section    | Line | KM Post | Segment     | Nearest Station | Annual GMT | Current GMT Passed | Radius (in m) | Degree Of Curve | Year of Laying | Length of Segment |
|---------------|------------|------|---------|-------------|-----------------|------------|--------------------|---------------|-----------------|----------------|-------------------|
| 1.1           | Section 7E | SL   | 65.63   | Sharp Curve | MONAPO          | 18         | 118.37             | 600           | 2.92            | 2015           | 547.00            |
| 1.2           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 1.3           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 1.4           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 1.5           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 1.6           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 2.1           | Section 7E | SL   | 111.69  | Tangent     | NAMIALO         | 18         | 118.37             | 0             | 0.00            | 2015           | 460.00            |
| 2.2           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 2.3           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 2.4           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 2.5           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 2.6           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 3.1           | Section 5  | SL   | 76.85   | Sharp Curve | LIWONDE         | 18         | 118.37             | 405           | 4.32            | 2015           | 291.50            |
| 3.2           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 3.3           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 3.4           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 3.5           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 3.6           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 4.1           | Section 5  | SL   | 72.78   | Mild Curve  | LIWONDE         | 18         | 118.37             | 1830          | 0.96            | 2015           | 1164.70           |
| 4.2           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 4.3           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 4.4           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 4.5           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 4.6           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 5.1           | Section 5  | SL   | 62.60   | Mild Curve  | MOLIPA          | 18         | 118.37             | 3660          | 0.48            | 2015           | 1059.00           |
| 5.2           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 5.3           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 5.4           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 5.5           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 5.6           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 6.1           | Section 7A | SL   | 485.88  | Tangent     | MUTUALI         | 18         | 118.37             | 0             | 0.00            | 2015           | 200.00            |
| 6.2           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |
| 6.3           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 6.4           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 6.5           |            |      |         |             |                 |            |                    |               |                 |                |                   |
| 6.6           |            |      | 2015    |             |                 |            |                    |               |                 |                |                   |





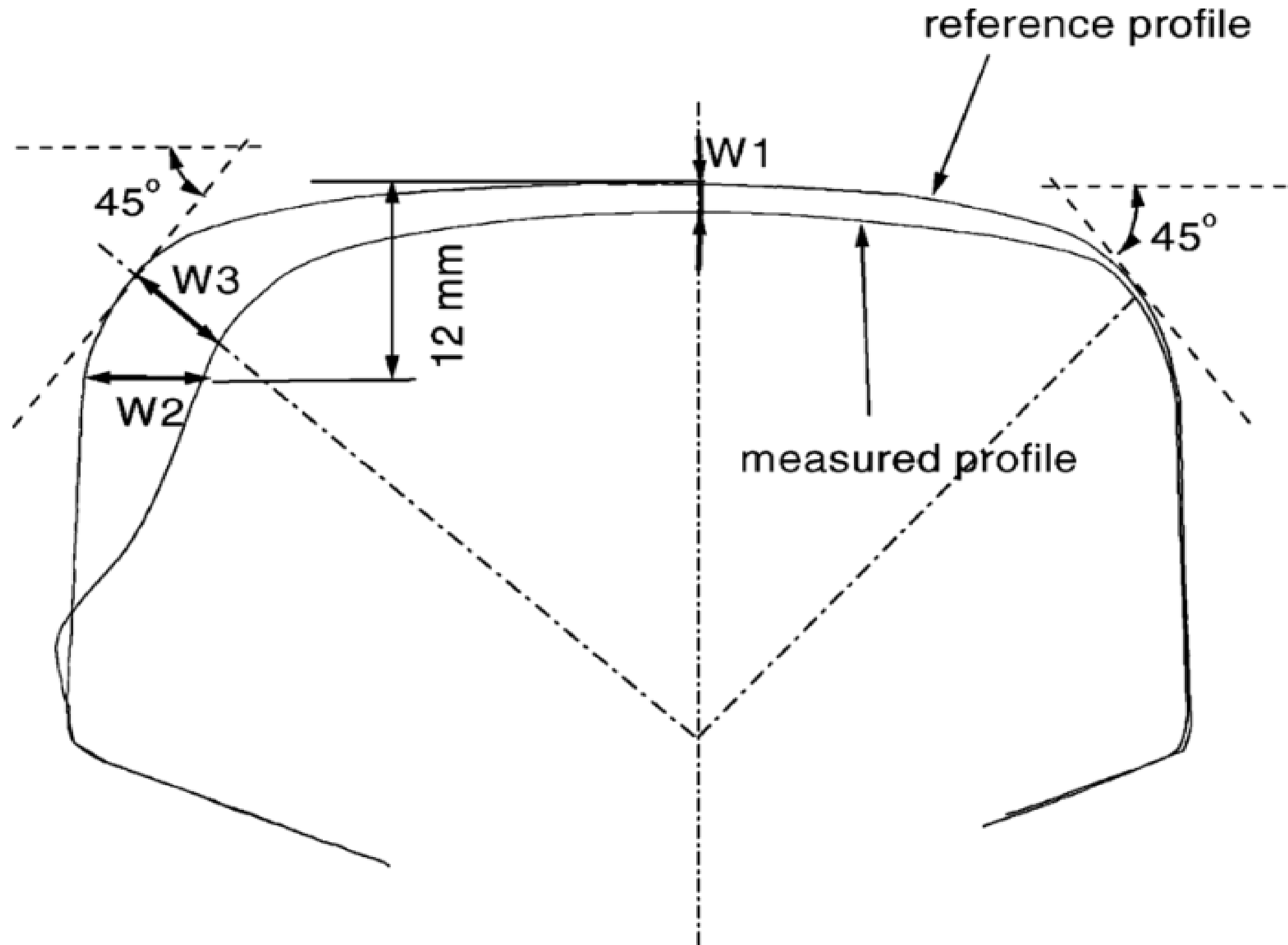
# Condition of Rail





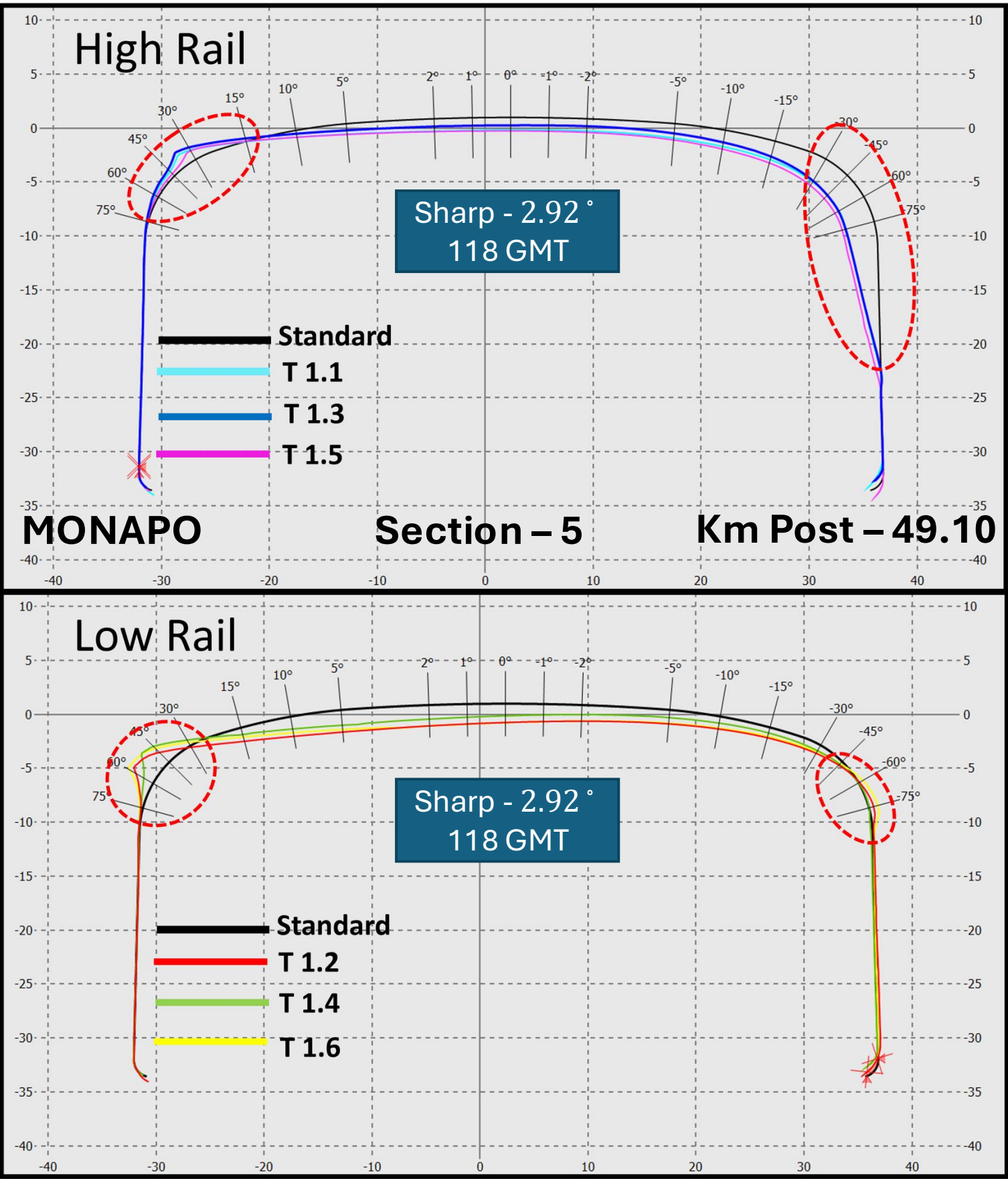


# Condition of Rail - Wear Values





# Condition of Rail



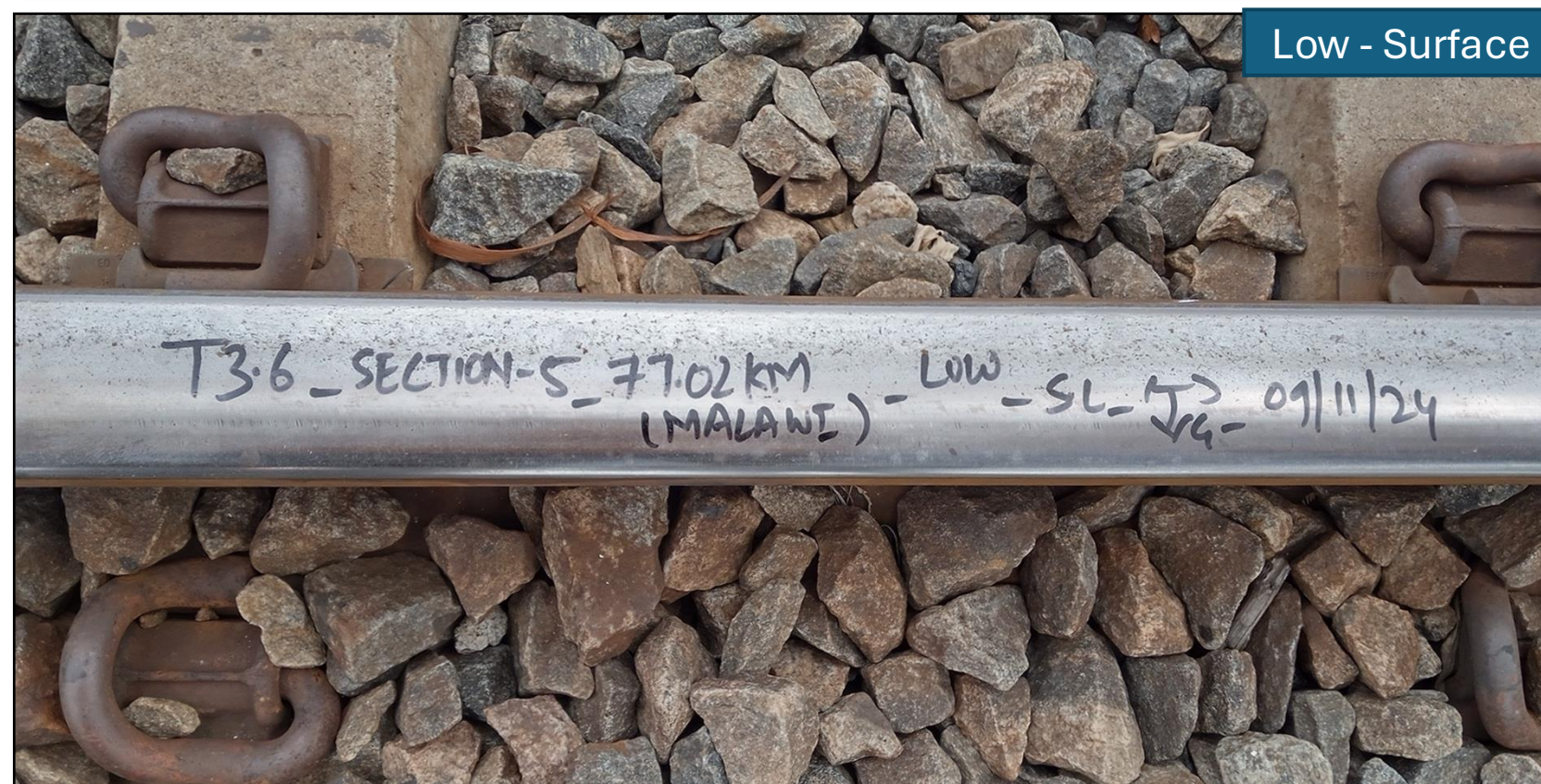
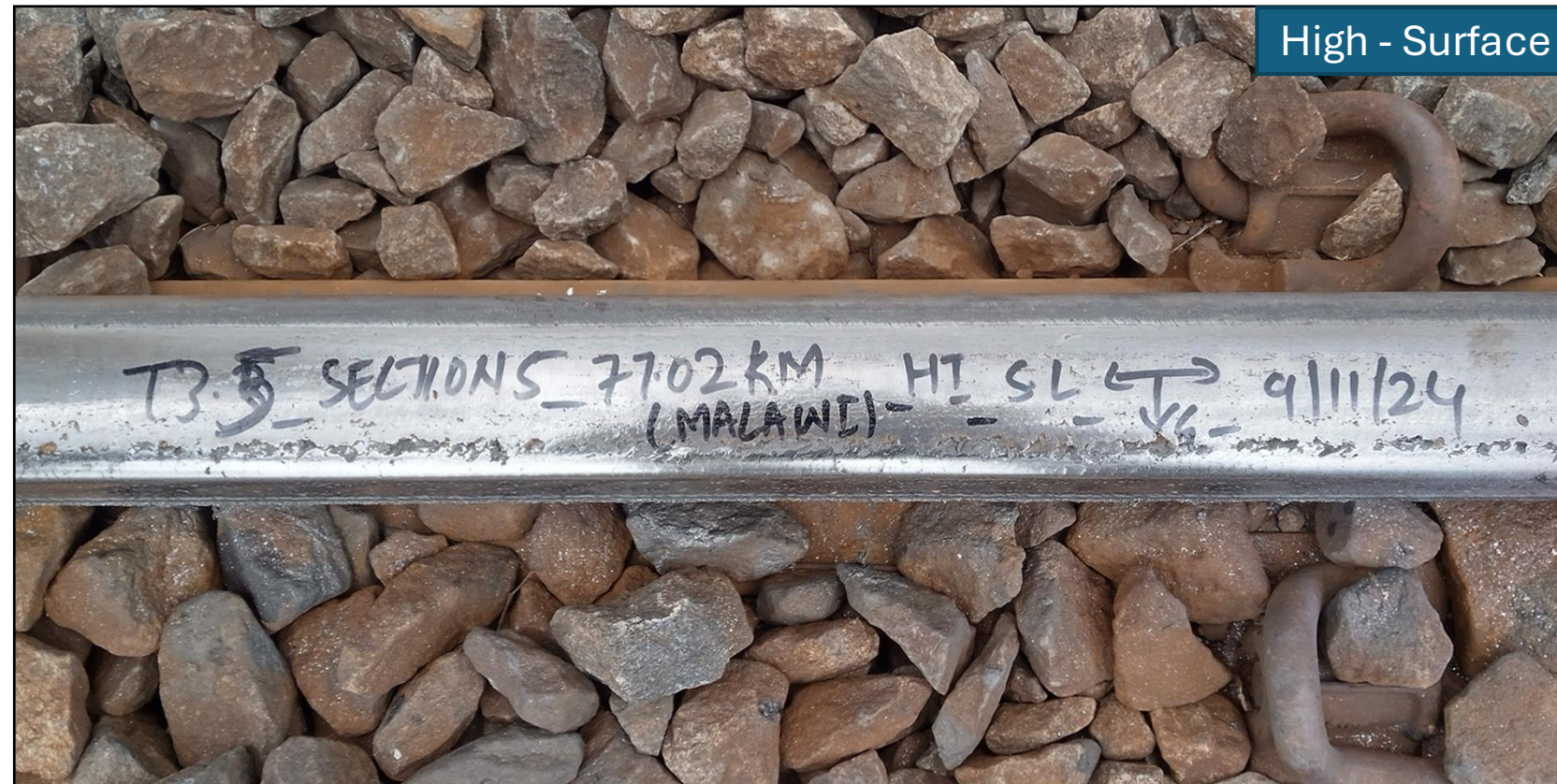
| Test Point | Rail | W1    | W2     | W3    |
|------------|------|-------|--------|-------|
| 1.1        | High | 1.345 | 2.107  | 2.697 |
| 1.3        |      | 0.920 | 2.026  | 2.707 |
| 1.5        |      | 1.438 | 2.399  | 3.131 |
| 1.2        | Low  | 1.928 | -0.185 | 0.585 |
| 1.4        |      | 1.311 | 0.095  | 0.459 |
| 1.6        |      | 1.936 | -0.117 | 0.385 |

- Excessive wear is observed on both rails.
- The low rail is predominantly experiencing vertical wear.
- The high rail is subjected to significant lateral wear.
- This level of wear is notable considering that only **118 GMT** has accumulated till Nov 2024.
- The trend indicates accelerated deterioration

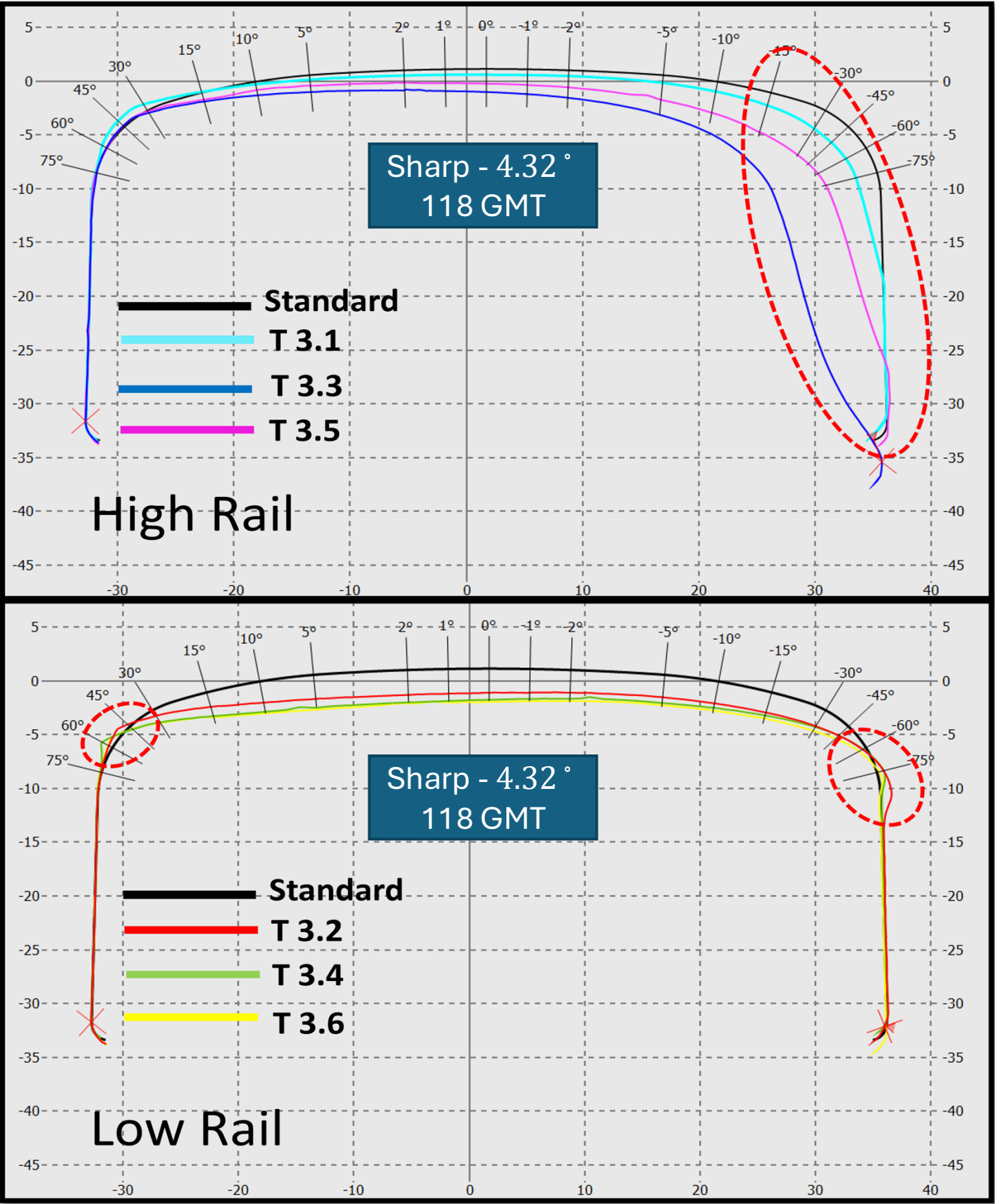




# Condition of Rail







| Test Point | Rail | W1    | W2     | W3    |
|------------|------|-------|--------|-------|
| 3.1        | High | 0.511 | 1.079  | 2.038 |
| 3.3        |      | 2.118 | 8.387  | 9.029 |
| 3.5        |      | 1.339 | 3.497  | 4.832 |
| 3.2        | Low  | 2.207 | -0.263 | 1.155 |
| 3.4        |      | 2.867 | -0.079 | 1.156 |
| 3.6        |      | 3.065 | 0.058  | 1.560 |

- Excessive wear is observed on both rails.
- The low rail is predominantly experiencing vertical wear.
- The high rail is subjected to significant lateral wear.
- This level of wear is notable considering that only **118 GMT** has accumulated till Nov 2024.
- The trend indicates accelerated deterioration





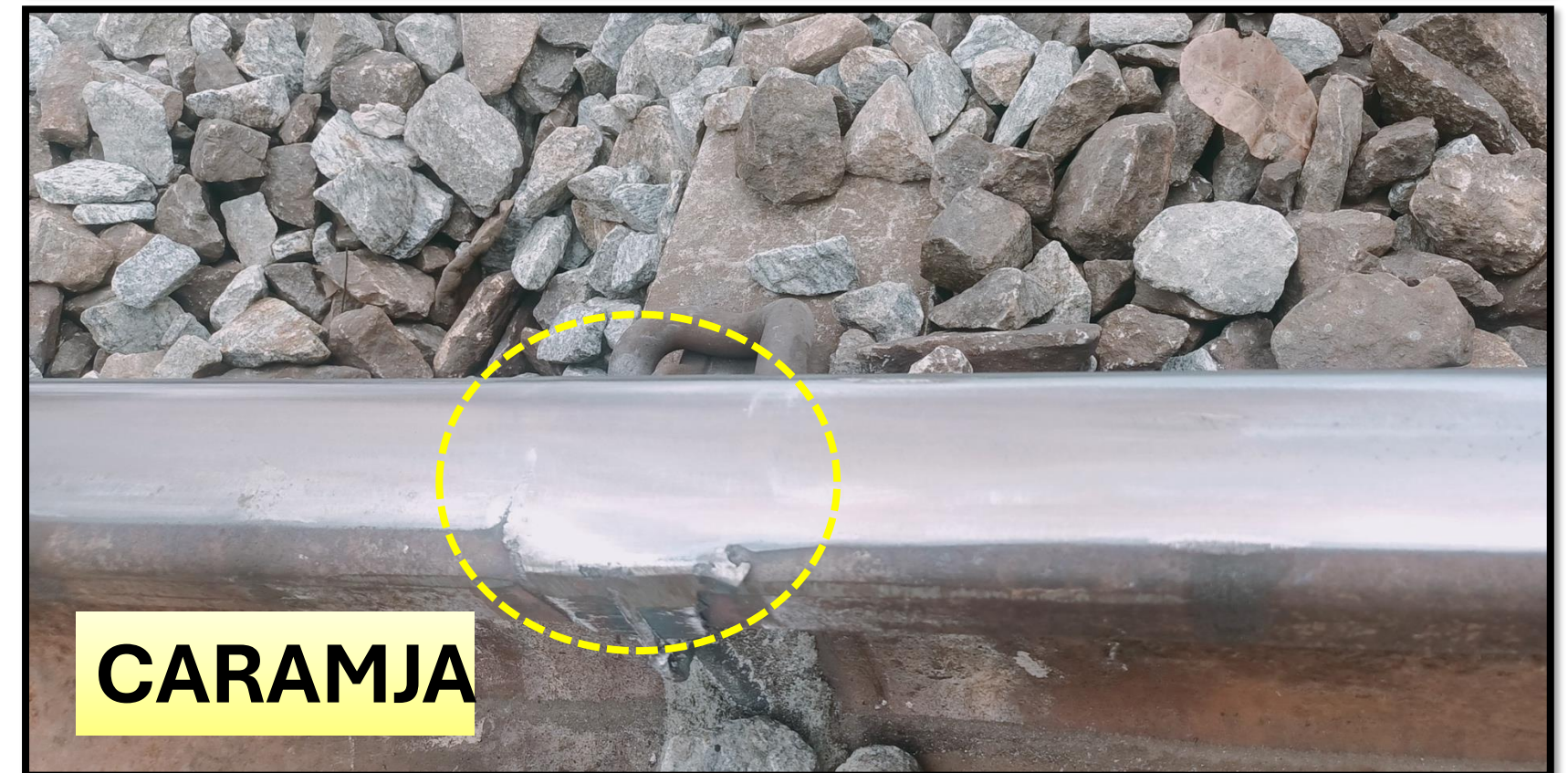
# Condition of Rail







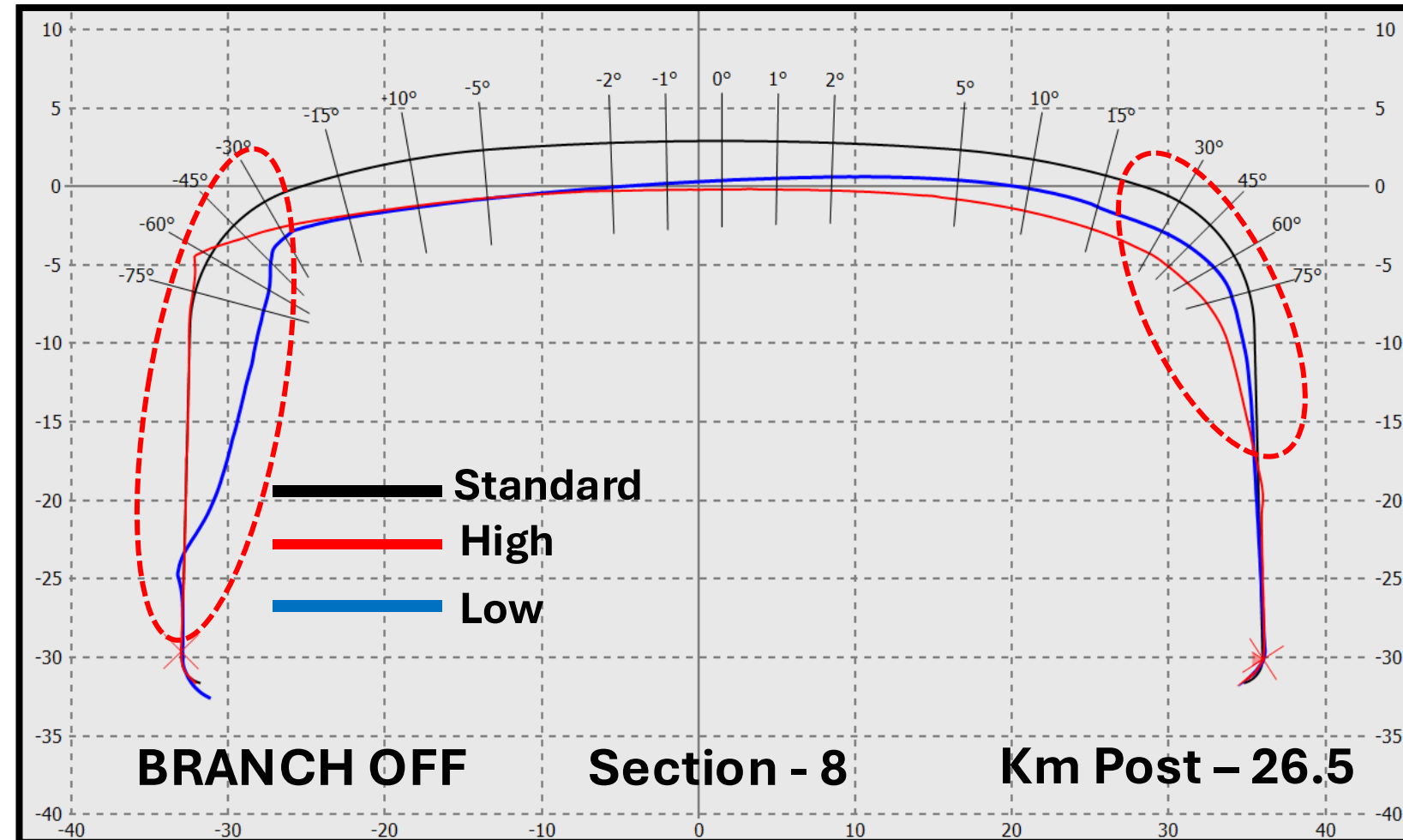
# Condition of Rail



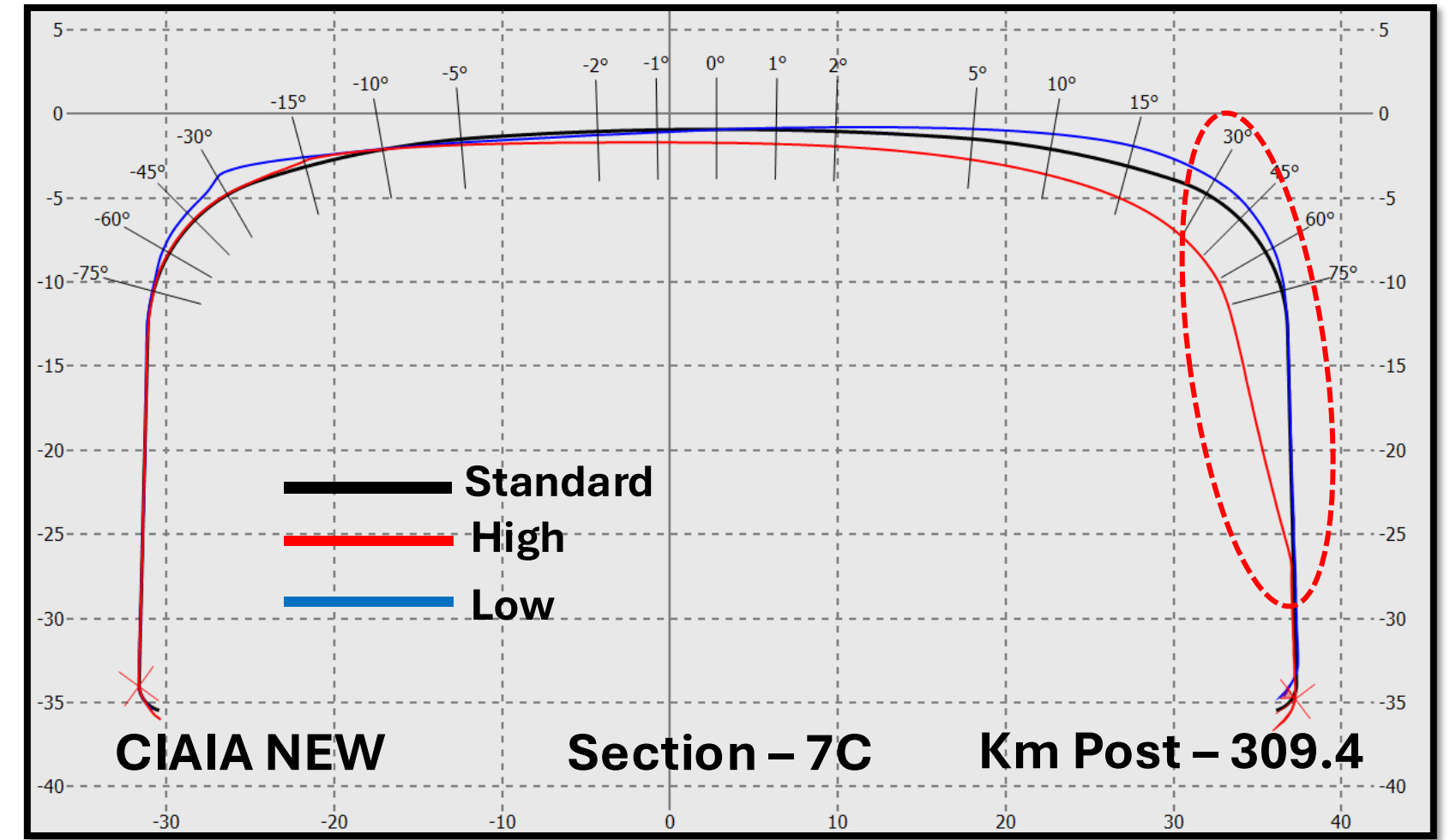




# Condition of Rail



| Location   | Rail | W1    | W2    | W3    |
|------------|------|-------|-------|-------|
| BRANCH OFF | High | 3.125 | 1.296 | 3.739 |
| BRANCH OFF | Low  | 2.543 | 0.453 | 1.908 |

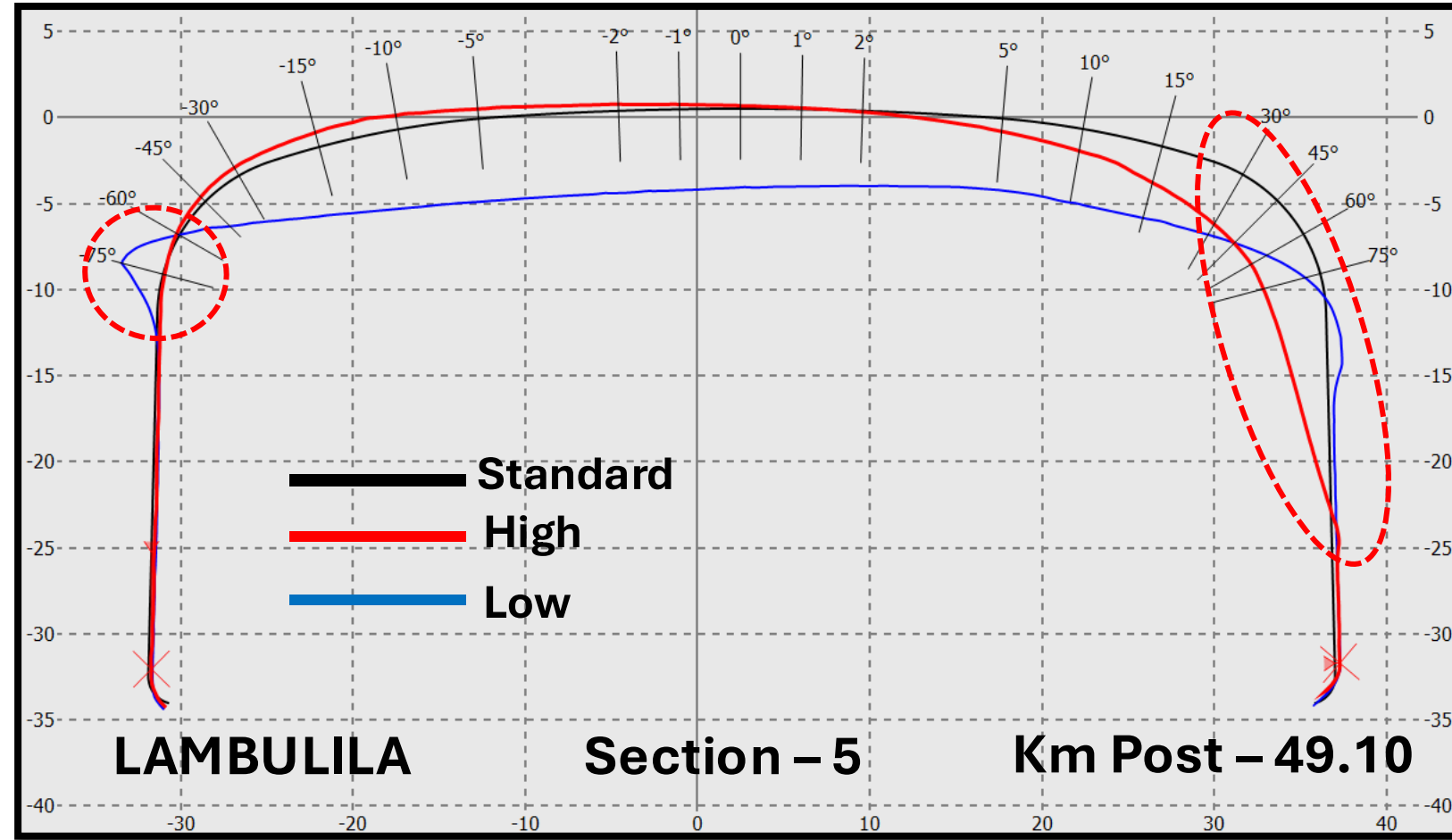


| Location  | Rail | W1    | W2     | W3     |
|-----------|------|-------|--------|--------|
| CIAIA NEW | High | 1.144 | 2.626  | 3.375  |
| CIAIA NEW | Low  | 0.369 | -0.093 | -0.653 |

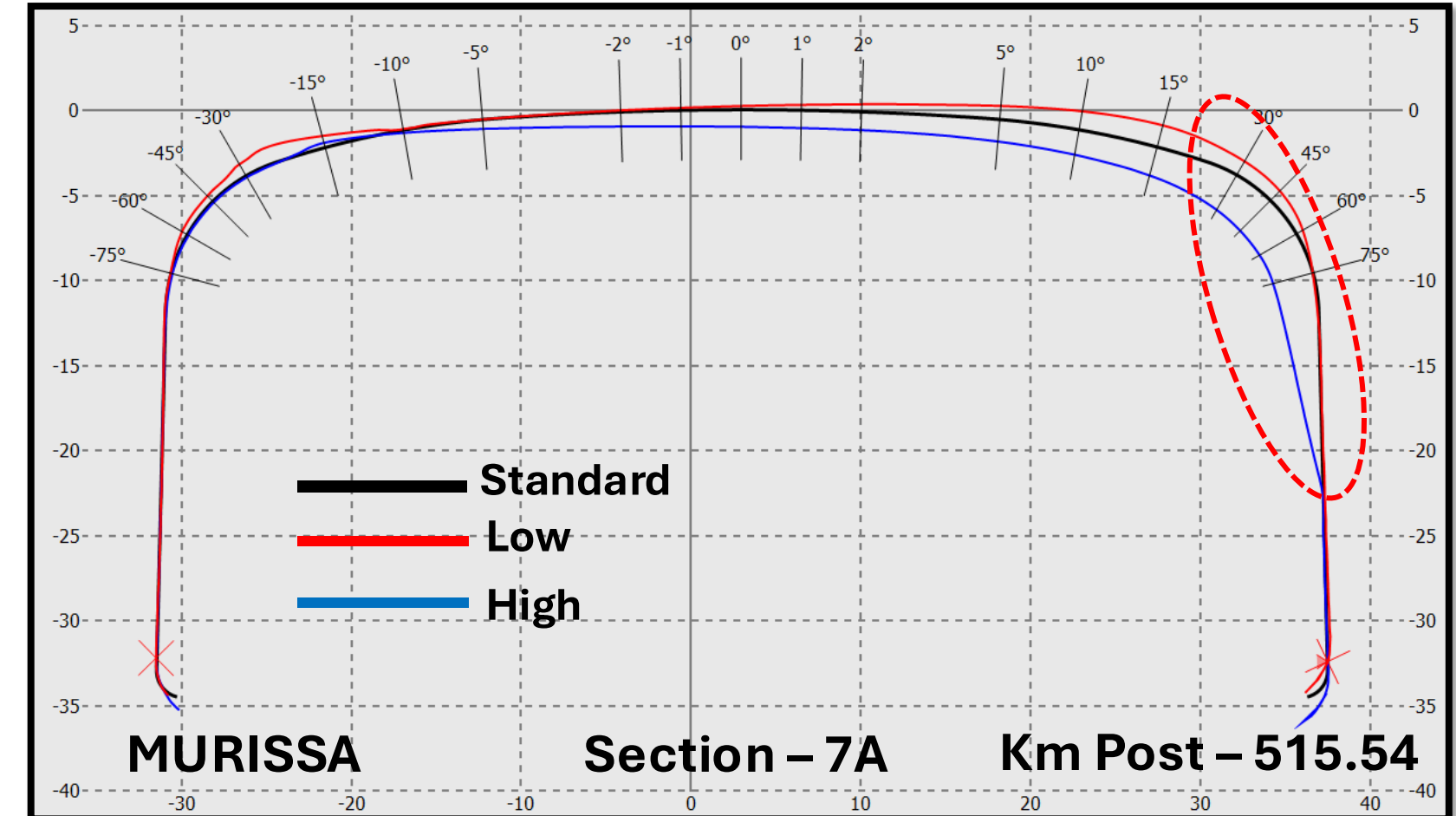




# Condition of Rail



| Location  | Rail | W1    | W2     | W3    |
|-----------|------|-------|--------|-------|
| LAMBULILA | High | 0.155 | 2.679  | 3.889 |
| LAMBULILA | Low  | 4.902 | -0.639 | 3.863 |

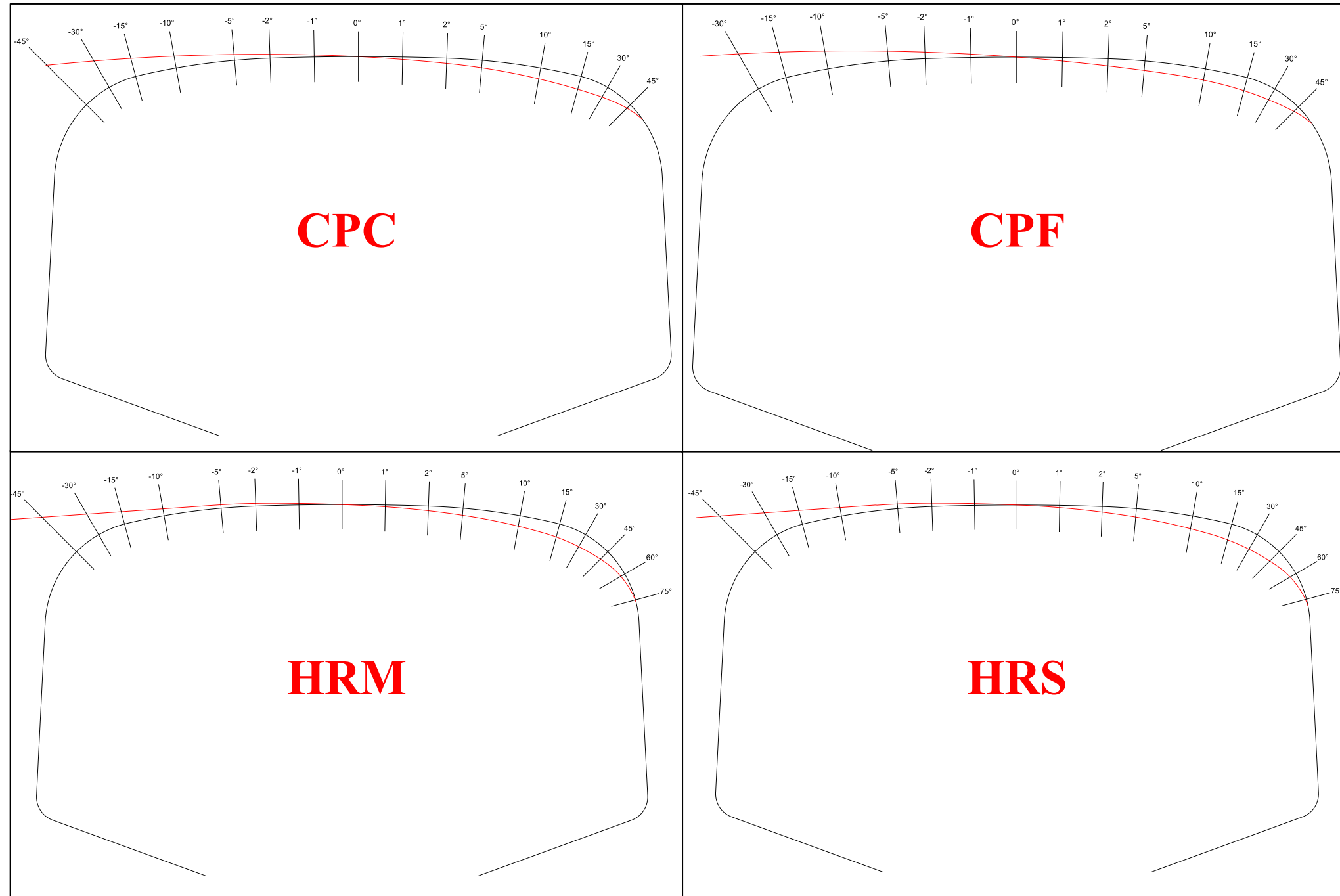


| Location | Rail | W1    | W2    | W3     |
|----------|------|-------|-------|--------|
| MURISSA  | High | 1.762 | 1.934 | 3.089  |
| MURISSA  | Low  | 0.536 | 0.040 | -0.233 |





# Rail Rehabilitation – move to Asymmetric Profiles



Templates designed – Asymmetric Rail Profiles

**Contact Point Centre (CPC):** This is implemented on the tangent track (Downline) and low side of mild curves. This will provide a tighter crown radius and appropriate gauge corner relief to achieve stable and consistent rail contact throughout the tangents.

**Contact Point Field (CPF):** This is implemented on the tangent track (Upline) and low side of Sharp curves. This will provide a tighter crown radius and appropriate gauge corner relief to achieve stable and consistent rail contact throughout the tangents ensuring truck hunting does not establish.

**High Rail Mild (HRM):** This template is implemented on the high side of the Mild curves. This provide gauge biased semi-conformal contact to promote steering and aid in reducing RCF development and growth.

**High Rail Sharp (HRS):** This template is implemented on the high side of the Sharp curves. This provide gauge biased semi-conformal contact to promote steering and aid in reducing RCF development and growth.





**Initial Corrective Strategy:** *The objective is to remove accumulated surface and sub-surface defects*

## Goals for Nacala:

1. Remove RCF cracks & plastic flow
2. Target gauge corner & running band
3. Depth based on DPT observations
4. Multi-pass reprofiling if required
5. Reset rail surface condition







# Rail Reprofilng based Rehabilitation

**Long-Term Preventive Strategy:** *The objective is to shift from corrective maintenance to condition based maintenance in a structured manner*

| Current Challenge | Delayed reprofiling → accelerated RCF growth<br>Non-uniform maintenance coverage                                                                                                |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposed Strategy | Define reprofiling intervals based on: <ul style="list-style-type: none"><li>➤ GMT accumulation</li><li>➤ Defect growth rate</li><li>➤ Track geometry (curve/tangent)</li></ul> |
| Cycle Design      | <ul style="list-style-type: none"><li>➤ High tonnage routes → shorter intervals</li><li>➤ Curves → higher frequency than tangents</li></ul>                                     |
| Monitoring Tools  | <ul style="list-style-type: none"><li>➤ MiniProf measurements</li><li>➤ Visual inspection trends</li><li>➤ Wear rate analysis</li><li>➤ Defect density tracking</li></ul>       |
| Outcome           | <ul style="list-style-type: none"><li>✓ Controlled wear evolution</li><li>✓ Extended rail life</li><li>✓ Reduced emergency interventions</li></ul>                              |





**RCF Control Strategy:** *The objective is to minimize initiation and propagation of RCF defects*

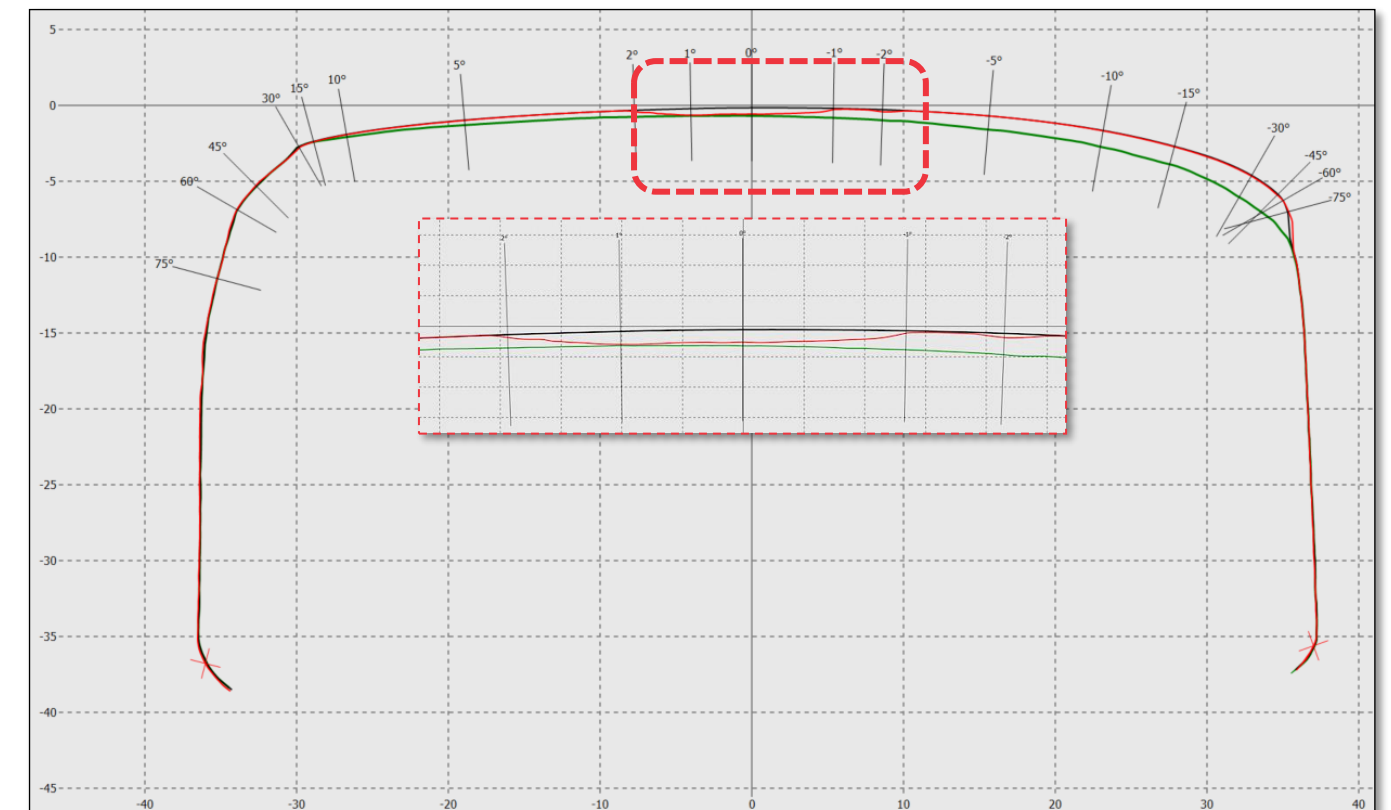
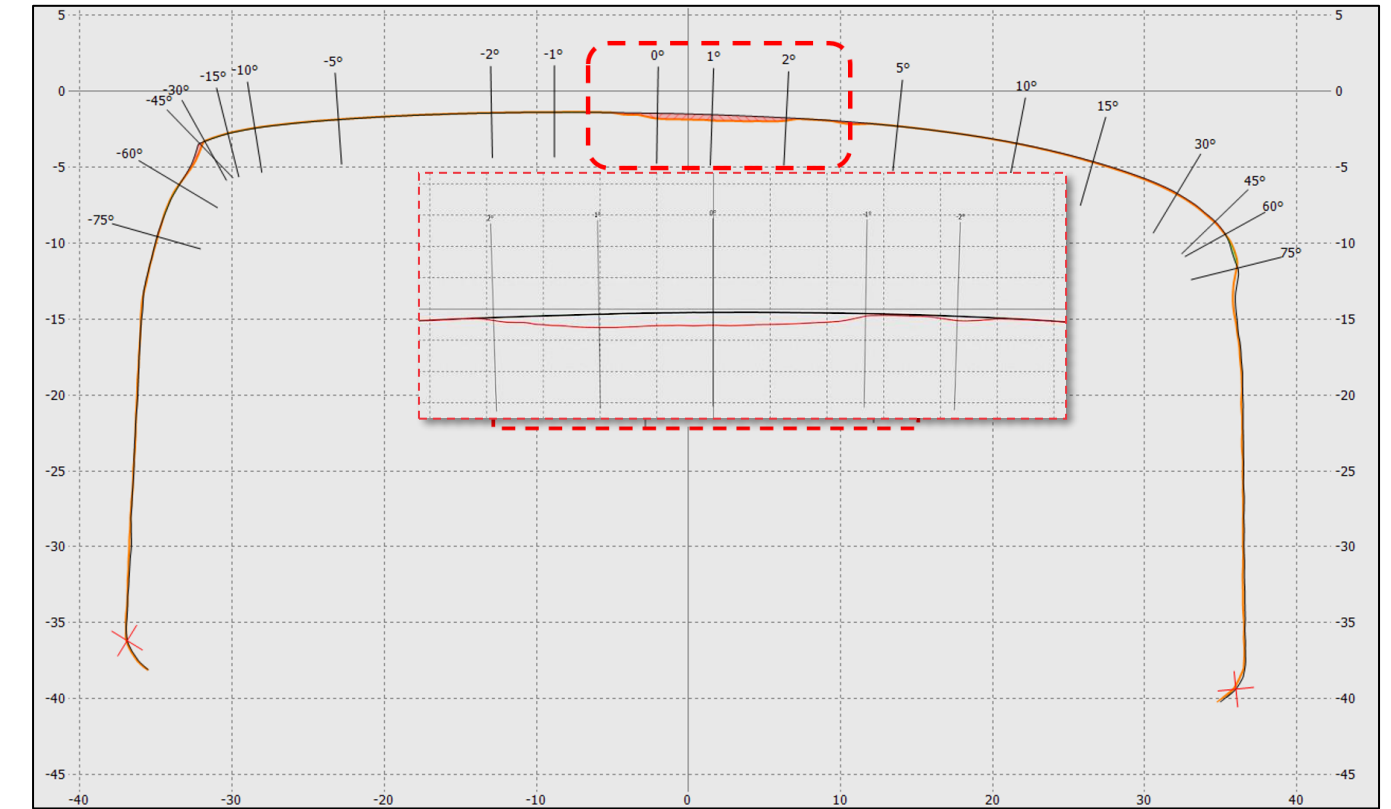
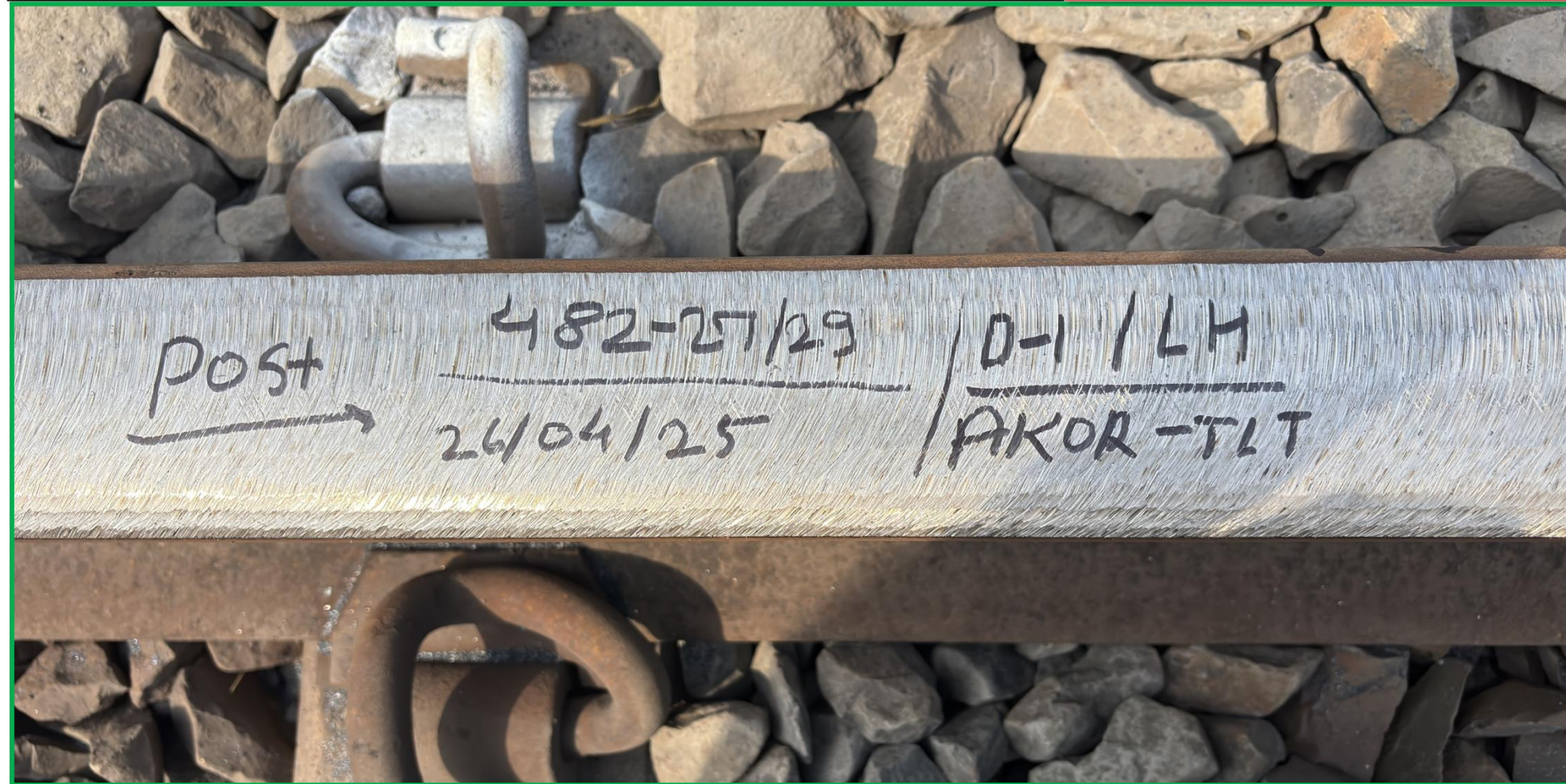
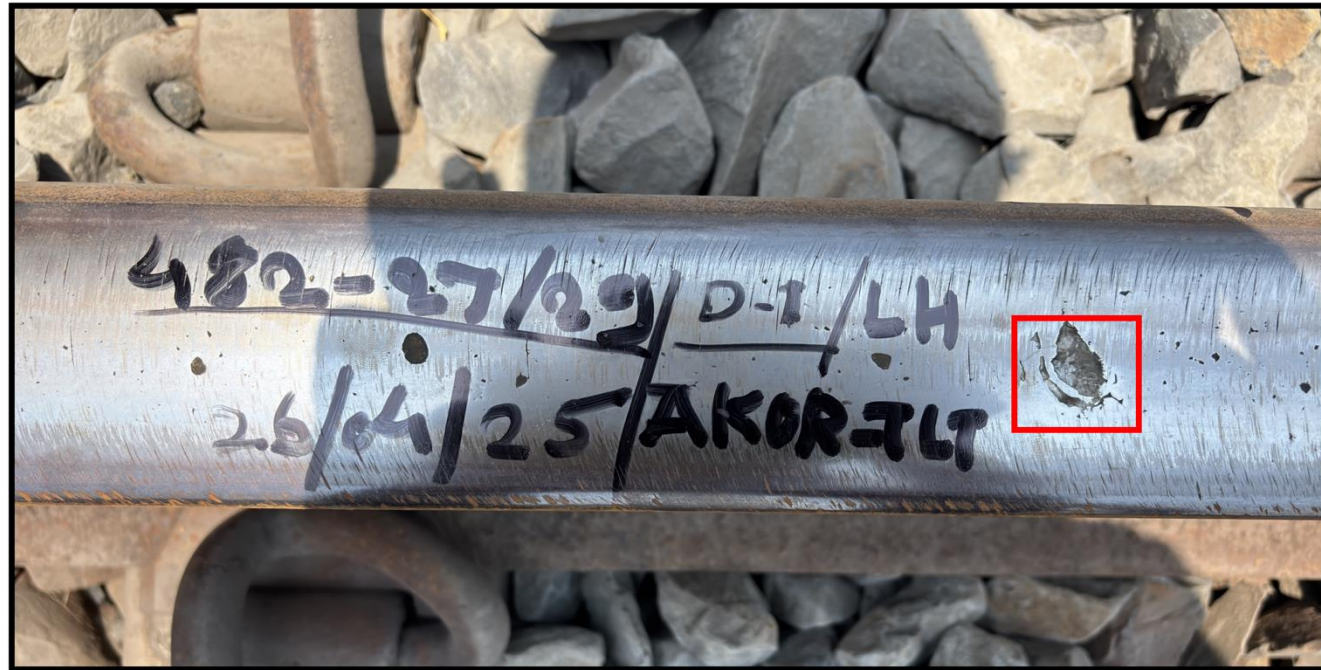
| Key Mechanisms   | <ul style="list-style-type: none"><li>➤ High contact stress at gauge corner</li><li>➤ Repeated cyclic loading</li></ul>                                                                                                                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control Measures | <p>Regular reprofiling to:</p> <ul style="list-style-type: none"><li>➤ Remove micro-cracks early</li><li>➤ Maintain compressive stress layer</li></ul> <p>Profile optimization:</p> <ul style="list-style-type: none"><li>➤ Improve contact distribution</li></ul> <p>Friction Management:</p> <ul style="list-style-type: none"><li>➤ Reduce friction and lateral forces</li></ul> |
| Strategy         | <ul style="list-style-type: none"><li>➤ To Interrupt crack growth cycle before critical depth</li><li>➤ To Maintain stable wheel–rail interaction</li></ul>                                                                                                                                                                                                                         |
| Outcome          | <ul style="list-style-type: none"><li>✓ Reduced crack density</li><li>✓ Lower risk of rail failure</li><li>✓ Improved operational reliability</li></ul>                                                                                                                                                                                                                             |





# Case Studies - Corrective Grinding

## Indian Railways – Central Railways (Zone)







# Case Studies – Vehicle dynamics

## Indian Railways – South Central Railways (Zone)



- The images represent the **pre and post-grinding contact band condition (IR - SCR Zone)**.
- In the pre-grinding condition, the contact band is predominantly observed on the **gauge face**, indicating unfavorable wheel–rail interaction.
- Post-grinding, the contact band has shifted towards the **Centre of the rail head**, reflecting a corrected rail profile.
- This shift is a direct outcome of **tangent track grinding**, which restores the desired rail geometry.
- Proper contact band positioning plays a crucial role in **maintaining stable vehicle dynamics and ensuring better wheel rail interaction**.
- Thus, grinding in SCR helps in optimizing wheel-rail contact, reducing lateral forces and improving overall ride quality and rail life.





# Review Table

| Option(s)                               | Advantages                                                                                                | Limitations for Nacala                                | Suitability                                                   |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------|
| <b>Preventive Rail Grinding</b>         | Low metal removal, good lifecycle control                                                                 | Requires already healthy rail profiles                | Medium at kick-off                                            |
| <b>Corrective Rail Grinding</b>         | Restores damaged rails quickly                                                                            | High metal removal, traffic disruption, reactive only | Medium as limited coverage                                    |
| <b>Preventive Gradual Rail Grinding</b> | Controlled profile restoration over multiple cycles, immediate implementation of Asymmetric rail profiles | Requires disciplined planning                         | Medium to High as this offered most flexibility               |
| <b>Milling</b>                          | Removes severe damage quickly, precise reprofiling                                                        | High metal removal, traffic disruption, reactive only | Low-Medium as long term goal is to move to preventive cycles. |

*The challenge was not whether reprofiling was required - it was selecting the right intervention strategy*





## Why Preventive Gradual Grinding Was Selected

- Existing rails show wear + RCF but remain recoverable
- Need to maintain traffic on heavy haul corridor
- Limited metal removal preserves rail life
- Progressive reprofiling reduces operational risk by initial move to asymmetric rail profile
- Suited for constrained maintenance windows
- Lower lifecycle cost than repeated corrective action or milling
- Scalable as tonnage rises from 30 MGT to 40 MGT





- The current rail condition indicates **advanced wear and the presence of multiple surface defect types** across the track.
- Rail wear levels are **significantly high**, with a diverse range of surface defects observed.
- This condition necessitates a **structured rail reprofiling rehabilitation plan** to systematically restore rail profile and health.
- Proposed strategy of **Preventive Gradual Rail Grinding** being adopted by Nacala.
- Such a strategy will help in **controlling defect initiation and growth**, ensuring sustained rail performance.
- Strategy also offered the **most flexibility** for Nacala with a single asset only available in the country.
- Similar approaches adopted in India have demonstrated **proven effectiveness in enhancing rail life, performance and overall track sustainability**.



